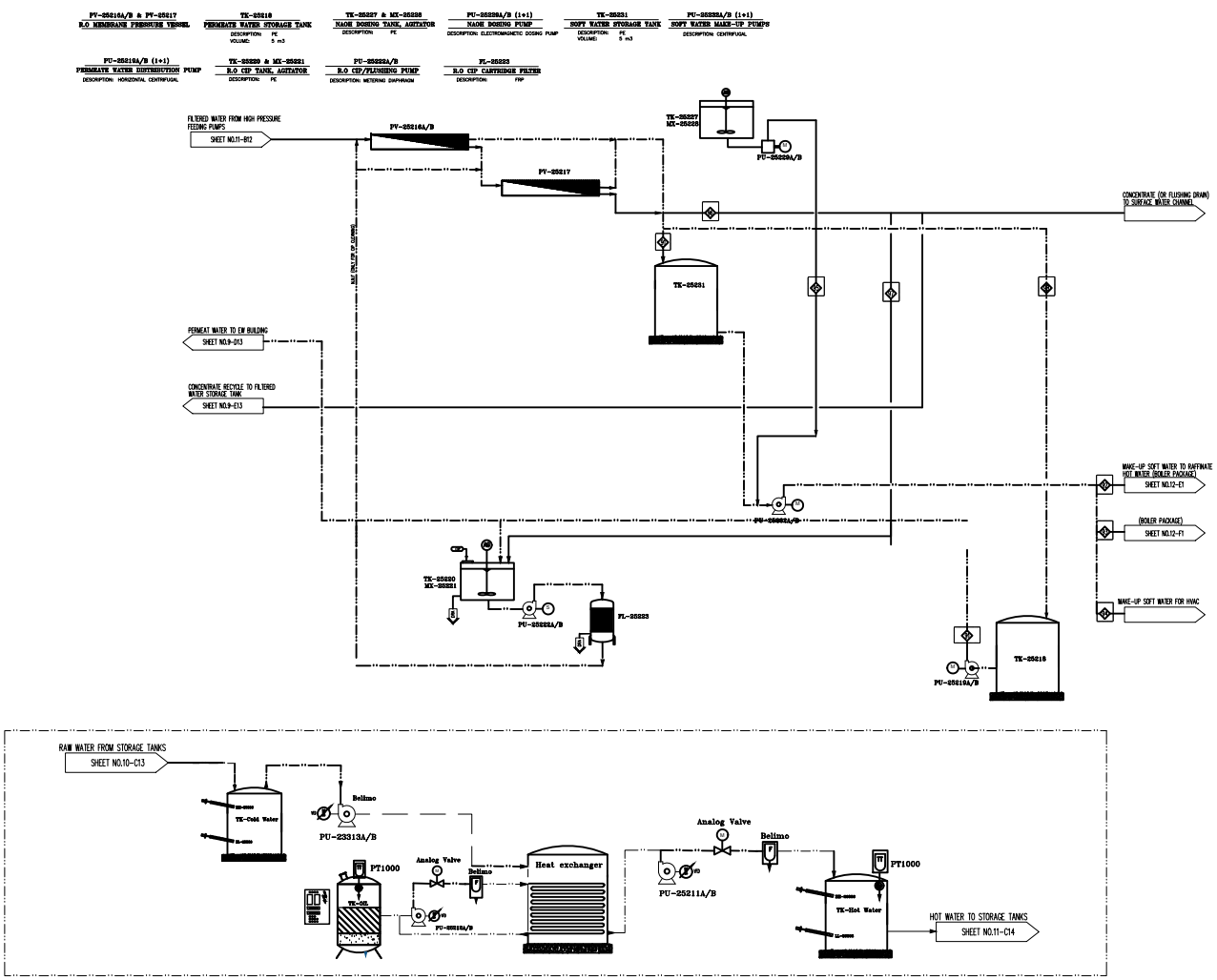

OPERATING PROCESS PARAMETERS (NOTE-2)

STREAM	DECRIPTION	FLOW RATE m^3/hr (AVERAGE)	FLOW RATE m^3/hr (DESIGN)	DENSITY (kg/m^3)	PRESSURE (barg)	TEMPERATURE (°C)	STREAM	DECRIPTION	FLOW RATE m^3/hr (AVERAGE)	FLOW RATE m^3/hr (DESIGN)	DENSITY (kg/m^3)	PRESSURE barg (DESIGN)	TEMPERATURE (°C)
91	WATER FROM MEMBRANE TO SOFT WATER STORAGE TANK	5.58	6.20	1000	0.20	5	95	NaOH Dosing	0.66 lit/hr	0.74 lit/hr	~1000	10 (Max)	5
92	MAKE UP SOFT WATER TO RAFFINATE HOT WATER (BOILER) GENERATION PACKAGE	3.12	3.43	1000	3~4.5	5	96	CONCENTRATE WATER FROM MEMBRANE	5.47	6.08	1000	19.72	5
93	MAKE UP SOFT WATER TO EW HOT WATER (BOILER) GENERATION PACKAGE	1	1.1	1000	3~4.5	5	97	CONCENTRATE WATER FROM MEMBRANE TO CIP TANK	2.25	2.50	1000	1	5
94	MAKE UP SOFT WATER FOR HVAC	1.5	1.65	1000	3~4.5	5	98	WATER FROM MEMBRANE TO PERMEATE WATER STORAGE TANK	1.93	2.15	1000	0.20	5

REFERENCE

REFERENCE. TITLE	REFERENCE. No.
PROCESS EQUIPMENT SIZING AND CALCULATION	11L1-CM-D-200-PR-CAL-1010
PFD SYMBOL & LEGEND	SHEET NO.1

NOTES

1. Package details will be finalized by vendor.
2. Permeate /Potable water consumption is finalized during detail design.
3. SMBS Dosing by CIP package should be considered.

REV.	DATE	DESCRIPTION	DESIGNED	CHECKED	APPROVED	AUTHORIZED
06	24-01-2021	ISSUED FOR CONSTRUCTION	Ah.Soesi	E.Zeinedini	SH.Kasaeian	A.Yousefi
05	04-05-2020	ISSUED FOR CONSTRUCTION	M.Kianersi	F.Delkhah	SH.Kasaeian	K.Ziarati
04	24-12-2019	ISSUED FOR CONSTRUCTION	M.Kianersi	F.Delkhah	SH.Kasaeian	K.Ziarati
03	09-11-2019	ISSUED FOR CONSTRUCTION	M.Kianersi	F.Delkhah	M.Borati	E.Rahimi
02	27-07-2019	ISSUED FOR APPROVAL	M.Kianersi	F.Delkhah	M.Borati	E.Rahimi
01	09-07-2019	ISSUED FOR APPROVAL	M.Kianersi	F.Delkhah	M.Borati	E.Rahimi
00	25-06-2019	ISSUED FOR REVIEW	M.K	F.D	M.B	E.R

Client: NATIONAL IRANIAN COPPER INDUSTRIES COMPANY (NICICO)

MC: PARMAN POOYESH CO.

Consultant: MEMSECO & ARIA HANGARD JOINT VENTURE

Contractor: CANY MEN CO.

Project: **SUNGUN Copper Ore Heap Leaching Plant Project**

Title: **Process Flow Diagram (PFD) and Mass Balance**
Plant / Instrument Air Generation & Distribution

Document Number:	DATE	SIZE	SCALE	SHEET NO.	REV. CLIENT	REV. VENDOR
11L1-CM-D-200-PR-PFD-1006	24-01-2021	A3	N.T.S	14 OF 14	06	--